

Examination Hall Seat Allocation Management System

Name:Amit Vishwakarma

Class:SYIT

Batch:1

RollNo: S057

```
import tkinter as tk
from tkinter import ttk, messagebox
import csv
import os
import re

# ----- SETTINGS -----
TOTAL_SEATS = 30
HALL_NO = "HALL-01"
FILE_NAME = "seat_allocations.csv"

students = []

# ----- VALIDATION -----
def valid_student_name(name):
    return bool(re.fullmatch(r"[A-Za-z ]{2,40}", name.strip()))

def valid_roll(roll):
    return bool(re.fullmatch(r"[A-Za-z0-9-]{1,15}", roll.strip()))

# ----- CSV FUNCTIONS -----
def save_allocations():
    with open(FILE_NAME, "w", newline="") as file:
        writer = csv.DictWriter(
            file,
            fieldnames=["name", "roll", "hall", "seat"]
        )
        writer.writeheader()
        writer.writerows(students)

def load_allocations():
```

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if not os.path.exists(FILE_NAME):
    return

try:
    with open(FILE_NAME, "r", newline="") as file:
        reader = csv.DictReader(file)
        for row in reader:
            if all(key in row for key in ("name", "roll", "hall", "seat")):
                students.append(row)
except (OSError, csv.Error):
    messagebox.showerror("Error", "Unable to load student data.")

```

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# ----- SEAT ALLOCATION -----
def get_next_seat():
    used_seats = {student["seat"] for student in students}

    for number in range(1, TOTAL_SEATS + 1):
        seat = f"SEAT-{number:02d}"
        if seat not in used_seats:
            return seat

    return None

```

```

def allocate_seat():
    name = name_entry.get().strip()
    roll = roll_entry.get().strip()

    if not valid_student_name(name):
        messagebox.showwarning(
            "Invalid Name",
            "Enter a valid name using letters only."
        )
    return

```

```

if not valid_roll(roll):
    messagebox.showwarning(
        "Invalid Roll Number",
        "Enter a valid roll number."
    )
    return

```

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if any(student["roll"].lower() == roll.lower()
        for student in students):
    messagebox.showwarning(

```

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        "Duplicate",
        "This roll number already exists."
    )
    return

seat = get_next_seat()

if seat is None:
    messagebox.showwarning(
        "Hall Full",
        "All seats have been allocated."
    )
    return

student = {
    "name": name,
    "roll": roll,
    "hall": HALL_NO,
    "seat": seat
}

students.append(student)

try:
    save_allocations()
except OSError as error:
    students.pop()
    messagebox.showerror("Error", str(error))
    return

refresh_table()
name_entry.delete(0, tk.END)
roll_entry.delete(0, tk.END)

messagebox.showinfo(
    "Success",
    f"Seat allocated successfully!\nSeat: {seat}"
)

# ----- TABLE FUNCTIONS -----
def refresh_table(data=None):
    for item in table.get_children():
        table.delete(item)

```

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records = students if data is None else data
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```
for student in records:
```

```
    table.insert(
        "",
        tk.END,
        values=(
            student["name"],
            student["roll"],
            student["hall"],
            student["seat"]
        )
    )
```

```
available_label.config(
    text=f"Available Seats: {TOTAL_SEATS - len(students)}"
)
```

```
def on_search():
```

```
    search_value = search_entry.get().strip().lower()
```

```
    if not search_value:
```

```
        refresh_table()
```

```
        return
```

```
    filtered = [
```

```
        student for student in students
```

```
        if search_value in student["name"].lower()
```

```
        or search_value in student["roll"].lower()
```

```
    ]
```

```
    refresh_table(filtered)
```

```
def on_show_all():
```

```
    search_entry.delete(0, tk.END)
```

```
    refresh_table()
```

```
# ----- DELETE SELECTED -----
```

```
def on_delete_selected():
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```
    selected = table.selection()
```

```
    if not selected:
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```
        messagebox.showwarning(
```

```
            "Delete Data",
```

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        "Please select a student first."
    )
    return

values = table.item(selected[0], "values")
roll = values[1]

confirm = messagebox.askyesno(
    "Confirm Delete",
    f"Are you sure you want to delete {values[0]}?"
)

if not confirm:
    return

index = next(
    (i for i, student in enumerate(students)
     if student["roll"] == roll),
    None
)

if index is None:
    messagebox.showerror("Error", "Student record not found.")
    return

deleted_record = students.pop(index)

try:
    save_allocations()
except OSError as error:
    students.insert(index, deleted_record)
    messagebox.showerror("Error", str(error))
    return

refresh_table()
messagebox.showinfo(
    "Deleted",
    "Selected student deleted successfully."
)

# ----- GUI -----
def create_app():
    global name_entry, roll_entry
    global search_entry, table, available_label

```

```

root = tk.Tk()
root.title("MVLU College - Examination Hall Seat Allocation")
root.geometry("850x600")
root.resizable(False, False)

title = tk.Label(
    root,
    text="MVLU COLLEGE",
    font=("Arial", 20, "bold")
)
title.pack(pady=(15, 0))

subtitle = tk.Label(
    root,
    text="EXAMINATION HALL SEAT ALLOCATION",
    font=("Arial", 14, "bold")
)
subtitle.pack(pady=5)

available_label = tk.Label(
    root,
    text=f"Available Seats: {TOTAL_SEATS}",
    font=("Arial", 12, "bold")
)
available_label.pack(pady=10)

# Student entry section
form_frame = ttk.LabelFrame(
    root,
    text="Student Details",
    padding=10
)
form_frame.pack(fill="x", padx=20, pady=10)

ttk.Label(form_frame, text="Student Name:").grid(
    row=0, column=0, padx=5, pady=5, sticky="w"
)

name_entry = ttk.Entry(form_frame, width=30)
name_entry.grid(row=0, column=1, padx=5, pady=5)

ttk.Label(form_frame, text="Roll Number:").grid(
    row=1, column=0, padx=5, pady=5, sticky="w"
)

```

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)

roll_entry = ttk.Entry(form_frame, width=30)
roll_entry.grid(row=1, column=1, padx=5, pady=5)

ttk.Button(
    form_frame,
    text="Allocate Seat",
    command=allocate_seat
).grid(row=0, column=2, rowspan=2, padx=20)

# Search section
search_frame = ttk.LabelFrame(
    root,
    text="Search Student",
    padding=10
)
search_frame.pack(fill="x", padx=20, pady=5)

search_entry = ttk.Entry(search_frame, width=30)
search_entry.pack(side="left", padx=5)

ttk.Button(
    search_frame,
    text="Search",
    command=on_search
).pack(side="left", padx=5)

ttk.Button(
    search_frame,
    text="Show All",
    command=on_show_all
).pack(side="left", padx=5)

ttk.Button(
    search_frame,
    text="Delete Selected",
    command=on_delete_selected
).pack(side="left", padx=5)

# Student table
table_frame = ttk.Frame(root)
table_frame.pack(
    fill="both",

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        expand=True,
        padx=20,
        pady=10
    )

columns = ("Name", "Roll Number", "Hall", "Seat")

table = ttk.Treeview(
    table_frame,
    columns=columns,
    show="headings",
    height=12
)

for column in columns:
    table.heading(column, text=column)
    table.column(column, width=180, anchor="center")

scrollbar = ttk.Scrollbar(
    table_frame,
    orient="vertical",
    command=table.yview
)

table.configure(yscrollcommand=scrollbar.set)

table.pack(side="left", fill="both", expand=True)
scrollbar.pack(side="right", fill="y")

root.mainloop()

# ----- START PROGRAM -----
load_allocations()
create_app()

```

Output Screenshot:

MVLU College - Examination Hall Seat Allocation

MVLU COLLEGE

EXAMINATION HALL SEAT ALLOCATION

Available Seats: 30

Student Details

Student Name:

Roll Number:

Allocate Seat

Search Student

Search

Show All

Delete Selected

Name	Roll Number	Hall	Seat
------	-------------	------	------

MVLU College - Examination Hall Seat Allocation

MVLU COLLEGE

EXAMINATION HALL SEAT ALLOCATION

Available Seats: 30

Student Details

Student Name:

Amit Vishwakarma

Roll Number:

S057

Allocate Seat

Search Student

Search

Show All

Delete Selected

Name	Roll Number	Hall	Seat
------	-------------	------	------

MVLU COLLEGE EXAMINATION HALL SEAT ALLOCATION

Available Seats: 29

Student Details

Student Name:

Roll Number:

Allocate Seat

Search Student

Search

Show All

Delete Selected

Success

Seat allocated successfully!
Seat: SEAT-01

OK

Name	Roll Number	HALL-01	SEAT-01
Amit Vishwakarma	S057		

MVLU COLLEGE EXAMINATION HALL SEAT ALLOCATION

Available Seats: 29

Student Details

Student Name:

Roll Number:

Allocate Seat

Search Student

Search

Show All

De

Confirm Delete

Are you sure you want to delete Amit Vishwakarma?

Yes No

Name	Roll Number	HALL-01	SEAT-01
Amit Vishwakarma	S057		

